

Water Potability Prediction — KNIME Workflow

Project Overview

This project uses Machine Learning techniques to classify water samples as **safe** (`is_safe = 1`) or **unsafe** (`is_safe = 0`) based on the concentration of chemical substances and contaminants.

The workflow compares five classification algorithms:

- Random Forest
- Tree Ensemble
- XGBoost
- Naive Bayes
- Logistic Regression

Each model is evaluated in two configurations: one using feature selection and one using dimensionality reduction through Principal Component Analysis (PCA). The pipeline includes exploratory data analysis, data cleaning, outlier removal, class balancing, stratified data splitting, hyperparameter optimization, and model evaluation using confusion matrices, ROC curves, AUC, and Lift Charts.

Requirements

- **KNIME Analytics Platform 5.4** or a compatible version
- KNIME extensions for Machine Learning, Statistics, Optimization, JavaScript Views, and XGBoost
- The dataset file `waterQuality1.csv`

When opening the workflow, KNIME may report missing extensions. Install the required extensions and restart KNIME before running the project.

How to Run the Workflow

1. Open KNIME Analytics Platform.
2. Import `WaterQuality_Project(1).knwf` through **File → Import KNIME Workflow**, or drag the file into the KNIME Explorer.
3. Download the *Water Quality* dataset from Kaggle and locate the CSV file containing the expected 21 columns, including the target variable `is_safe`.
4. Open the configuration of the **CSV Reader (#70)** node. The workflow contains a local file path from the original development environment, so manually select your own `waterQuality1.csv` file.
5. Check the CSV preview and verify that the separator, column names, and data types are correctly recognized. Confirm the configuration.
6. Reset the workflow using **Reset Workflow** to remove any previously stored execution results.
7. Run the complete pipeline using **Execute All** or the green execution button.

The full execution may take several minutes because the workflow includes stratified cross-validation and hyperparameter optimization.

Results

The final outputs are available in the **Models**, **NO PCA**, **ROC Curve**, and **Lift Charts** sections. Open the **Scorer** node views to inspect accuracy, Cohen's Kappa, and confusion matrices. ROC and Lift Chart views can be used to compare model performance.

In the original project, **Random Forest without PCA** was selected as the preferred model because it produced the lowest number of false positives. This is especially important in water safety applications, where unsafe water must not be incorrectly classified as potable.

Reproducibility

The main randomized operations use seed **123**. To reproduce the original results, do not change the random seed, train/test split, class-balancing settings, or hyperparameter optimization configuration.